ADVANCED APPLICATION DEVELOPMENT (CSE 214)

Final Project Assignment

E-Commerce Analytics Platform with Multi-Agent Text2SQL AI Chatbot

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Technology Stack

- **Backend:** Spring Boot
- **Frontend:** Angular
- **Database:** MySQL / PostgreSQL
- **AI Integration:** Multi-Agent Text2SQL Agentic AI with LangGraph and Chainlit

Group Size: 2 students per group

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1. Project Overview

This project requires students to build a comprehensive e-commerce data analytics platform inspired by modern business intelligence tools like Metabase. The platform will serve three distinct user types with role-based access control and feature an AI-powered multi-agent Text2SQL chatbot that converts natural language questions into database queries.

1.1 Learning Objectives

- Design and implement a relational database schema by integrating multiple data sources
- Build RESTful APIs using Spring Boot with proper authentication and authorization
- Develop a responsive Angular frontend with dynamic data visualization
- Implement role-based access control (RBAC) for multi-tenant applications
- Build a Multi-Agent Text2SQL Chatbot using LangGraph and Chainlit, leveraging Agentic AI for e-commerce analytics

1.2 User Roles

Role	Responsibilities
Admin	System administration, store management (open/close), user management, platform analytics, and global configuration
Corporate User	Store management, inventory control, sales analytics, order fulfillment, customer insights, and revenue reporting
Individual User	Browse products, place orders, track shipments, view purchase history, write reviews, and access personal spending analytics

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2. Dataset Integration Guide

Students will work with **six Kaggle datasets** to construct a unified relational database. The goal is to merge, transform, and normalize these datasets into a coherent schema that supports all platform features.

2.1 Source Datasets

#	Dataset Name	Key Fields	Usage in Project
1	**E-Commerce Sales Forecast (UCI Online Retail)**	InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country	Orders, OrderItems, Products tables
2	**E-Commerce Customer Behavior**	CustomerID, Gender, Age, City, MembershipType, TotalSpend, ItemsPurchased, AvgRating, DiscountApplied, SatisfactionLevel	Users, CustomerProfiles tables
3	**E-Commerce Shipping Data**	ID, WarehouseBlock, ModeOfShipment, CustomerCareCalls, CustomerRating, CostOfProduct, PriorPurchases, ProductImportance, DiscountOffered	Shipments table
4	**E-Commerce Sales (Amazon)**	OrderID, Date, Status, Fulfilment, SalesChannel, ShipServiceLevel, Style, SKU, Category	Orders status and fulfilment fields
5	**Pakistan E-Commerce Orders**	ItemID, Status, CreatedAt, SKU, Price, QtyOrdered, GrandTotal, IncrementID, CategoryName, PaymentMethod	Orders, OrderItems, PaymentMethods
6	**Amazon US Customer Reviews**	Marketplace, CustomerID, ReviewID, ProductID, ProductTitle, ProductCategory, StarRating, HelpfulVotes, TotalVotes	Reviews table and Sentiment Analysis

2.2 Data Integration Matrix

Table	DS1: Online Retail	DS2: Customer Behavior	DS3: Shipping	DS4: Amazon Sales	DS5: Pakistan	DS6: Reviews
Users	✓ CustomerID	✓✓ Demographics	Gender	-	-	✓ CustomerID
Products	✓✓ StockCode, Desc	-	Cost	SKU	SKU	Title
Orders	✓✓ Invoice, Date	-	-	Status	✓✓ Full Order	-
Shipments	-	-	✓✓ Full Data	Service Level	-	-
Reviews	-	Satisfaction	Rating	-	-	✓✓ Full Reviews

2.3 Database Design Guidelines

Students must create a normalized relational schema by mapping and integrating the source datasets.

Core Entities:

- **Users:** Unified user table with role discrimination (admin, corporate, individual)
- **Stores:** Store information linked to corporate users
- **Products:** Product catalog with SKU, description, category, and pricing
- **Orders:** Order header with status, date, customer, and payment information
- **OrderItems:** Line items linking orders to products with quantity and price
- **Shipments:** Shipping details including warehouse, mode, and tracking
- **Reviews:** Customer reviews with ratings and helpfulness scores
- **Categories:** Product category hierarchy
- **CustomerProfiles:** Extended customer data including membership, demographics, and behavior metrics

Recommended Schema:

```
USERS (id, email, password_hash, role_type, gender)
    ├── STORES (id, owner_id, name, status)
    ├── CUSTOMER_PROFILES (id, user_id, age, city, membership_type)
    └── ORDERS (id, user_id, store_id, status, grand_total)
            ├── ORDER_ITEMS (id, order_id, product_id, quantity, price)
            └── SHIPMENTS (id, order_id, warehouse, mode, status)

PRODUCTS (id, store_id, category_id, sku, name, unit_price)
    └── REVIEWS (id, user_id, product_id, star_rating, sentiment)

CATEGORIES (id, name, parent_id)
```

2.4 Data Transformation Requirements (ETL)

- **Field Mapping:** Create mapping documentation showing how source fields map to target schema
- **Data Cleansing:** Handle missing values, duplicates, and inconsistent formats
- **ID Generation:** Generate surrogate keys for unified entities
- **Date Normalization:** Standardize all date formats to ISO 8601
- **Currency Handling:** Normalize prices to a single currency with exchange rate metadata

3. Technical Requirements

3.1 Backend (Spring Boot)

1. **RESTful API Design:** Implement CRUD operations for all entities following REST conventions
2. **Authentication:** JWT-based authentication with refresh token support
3. **Authorization:** Role-based access control using Spring Security
4. **Data Layer:** JPA/Hibernate with support for both MySQL and PostgreSQL
5. **Query Execution:** Dynamic query builder for chatbot-generated SQL
6. **API Documentation:** Swagger/OpenAPI specification
7. **Error Handling:** Global exception handler with meaningful error responses

3.2 Frontend (Angular)

1. **Responsive Design:** Mobile-first approach with adaptive layouts
2. **Component Architecture:** Modular components with lazy loading
3. **State Management:** NgRx or service-based state management
4. **Data Visualization:** Interactive charts using Chart.js, D3.js, or ngx-charts
5. **Chat Interface:** Real-time chat UI for AI assistant interaction
6. **Dynamic Dashboards:** Configurable widgets for analytics display

3.3 Database Requirements

- Support both MySQL and PostgreSQL (configurable via application properties)
- Create indexes for frequently queried fields

4. Feature Specifications by Role

4.1 Individual User Features

- Product browsing with search, filter, and sort capabilities
- Shopping cart management and checkout process
- Order placement with multiple payment method support
- Real-time order tracking and shipment status
- Purchase history with filtering and export
- Product review and rating submission
- Personal spending analytics dashboard
- Profile management and preferences

4.2 Corporate User Features

- Store dashboard with KPI overview
- Product catalog management (CRUD operations)
- Inventory tracking and low-stock alerts
- Order management and fulfillment workflow
- Sales analytics with customizable date ranges
- Customer segmentation and behavior analysis
- Revenue reports with drill-down capability
- Review management and response system

4.3 Admin Features

- Platform-wide analytics dashboard
- User management (create, suspend, delete accounts)
- Store approval and management (open/close stores)
- Category and taxonomy management
- System configuration and settings
- Audit logs and activity monitoring
- Cross-store comparison reports

5. Multi-Agent Text2SQL AI Chatbot

5.1 Introduction and Problem Statement

Not very long ago, only a handful of people (Data Engineers, Analysts) could query the SQL databases in an organization. The people who needed it the most—Sales and Marketing teams—didn't have the skills to query the databases. They had to rely on the technical teams to get information.

This multi-agent Text2SQL chatbot removes that dependency. It empowers all people in the organization to ask questions in natural English language.

The Challenge:

- Technical Barrier: Non-technical stakeholders cannot query databases directly
- SQL Complexity: Writing complex JOINs and aggregations requires expertise
- Time Consumption: Back-and-forth with data teams slows decision-making
- Visualization Gap: Raw query results don't provide immediate insights
- Error Prone: Manual SQL queries are susceptible to syntax and logical errors

The Solution: A conversational AI system that:

- Accepts natural language questions in plain English
- Validates scope to prevent irrelevant queries
- Generates optimized SQL queries automatically
- Handles errors intelligently with retry mechanisms
- Provides natural language explanations of results
- Creates visualizations when beneficial
- Streams execution process for transparency

5.2 Tech Stack for AI Component

- **Core Framework:** LangGraph (state machine architecture)
- **Frontend:** Angular (chat interface)
- **AI & LLM:** OpenAI API (gpt-4o-mini) or alternative LLM providers (Anthropic Claude, Ollama for local deployment)
- **Visualization:** Plotly (LLM-Generated Visualization Code)
- **Utilities:** python-dotenv, Pandas

5.3 Multi-Agent Architecture

The chatbot uses a multi-agent architecture where specialized AI agents work together to provide accurate, context-aware responses.

Agent Definitions:

Agent	Role	Responsibility
Guardrails Agent	Security and Scope Manager	Filters questions to ensure they are relevant to e-commerce data analysis, identifies greetings, and blocks out-of-scope queries
SQL Agent	SQL Expert	Generates only valid SQL queries without any formatting or explanation based on the database schema
Analysis Agent	Data Analyst	Explains database query results in natural language with clear insights
Visualization Agent	Visualization Specialist	Generates clean, executable Plotly code for data visualization when beneficial
Error Agent	Error Recovery Specialist	Diagnoses and fixes SQL errors with expert knowledge of database schemas

5.4 Agentic Flow

The system follows this workflow:

1. **User Input:** User asks a question in natural language
2. **Guardrails Check:** Guardrails Agent validates if the question is in scope
    - If greeting → sends friendly welcome message and ends
    - If out-of-scope → returns predefined rejection message
    - If in-scope → continues to SQL generation

3. **SQL Generation:** SQL Agent converts natural language into valid SQL query
4. **Query Execution:** System executes SQL safely against the database
5. **Error Handling:** If error occurs, Error Agent diagnoses and attempts to fix
6. **Analysis:** Analysis Agent explains results in natural language
7. **Visualization:** If beneficial, Visualization Agent generates Plotly charts
8. **Response:** Final answer with explanation and optional visualization returned to user

5.5 Agent State Management

Using TypedDict for state management ensures type safety and provides benefits:

- Python will warn if wrong data types are used
- Editor provides autocomplete and reduces mistakes
- State includes: question, sql_query, query_result, error, final_answer, visualization_code, is_in_scope, iteration_count

5.6 Agent Configurations Example

```
AGENT_CONFIGS = {
    "guardrails_agent": {
        "role": "Security and Scope Manager",
        "system_prompt": "You are a strict guardrails system that filters
questions
                          to ensure they are relevant to e-commerce data analysis."
    },
    "sql_agent": {
        "role": "SQL Expert",
        "system_prompt": "You are a senior SQL developer specializing in e-
commerce
                          databases. Generate only valid SQL queries without any
                          formatting or explanation."
    },
    "analysis_agent": {
        "role": "Data Analyst",
        "system_prompt": "You are a helpful data analyst that explains database
                          query results in natural language with clear insights."
    },
    "viz_agent": {
        "role": "Visualization Specialist",
        "system_prompt": "You are a data visualization expert. Generate clean,
                          executable Plotly code without markdown formatting."
    },
    "error_agent": {
        "role": "Error Recovery Specialist",
        "system_prompt": "You diagnose and fix SQL errors with expert knowledge
                          of database schemas and query optimization."
    }
}
```

5.7 Example Interactions

User Query (Natural Language)	Expected Chatbot Response
"Show me sales by category for last month"	Generate bar chart showing category sales with date filter applied
"What are my top 5 customers by revenue?"	Display ranked table or horizontal bar chart of top customers
"Compare this month vs last month"	Show comparison chart with period-over-period metrics
"Which products have the lowest ratings?"	Return sorted list of products by average rating
"What's the trend in order cancellations?"	Generate time series line chart of cancellation rates
"How many orders were shipped by air?"	Return count with natural language explanation

5.8 Role-Based Chatbot Access

The chatbot must respect data segregation principles based on user roles:

Role	Data Access Scope
Individual	Own orders, purchases, reviews, and spending patterns only
Corporate	Own store's products, orders, customers, reviews, and sales data
Admin	Full platform access including all stores, users, and aggregate data

5.9 Integration with Spring Boot

The AI chatbot service (Python/LangGraph) communicates with the Spring Boot backend via REST API or message queue. Students should implement:

- Dedicated endpoint for chat requests (/api/chat/ask)
- Query execution service that runs generated SQL safely
- Result formatting and passing to visualization agent
- Session management for conversation context

6. Project Deliverables

6.1 Required Submissions

1. **Source Code Repository:** GitHub repository with proper branching and commit history demonstrating collaboration between team members
2. **Technical Report (10-15 pages):**
    - Architecture decisions and justifications
    - Database design with ER diagrams
    - ETL process documentation with field mappings
    - API documentation overview
    - AI chatbot architecture explanation
    - Challenges faced and solutions implemented
3. **Live Project Presentation (10 minutes):**
    - Demonstrate all three user roles
    - Show AI chatbot functionality with various query types
    - Explain technical architecture

7. Grading Criteria

Criteria	Weight	Key Evaluation Points
Database Design & Data Integration	20%	Normalization, relationships, ETL quality
Backend Implementation (Spring Boot)	25%	API design, security, code quality
Frontend Implementation (Angular)	20%	UI/UX, responsiveness, visualization
AI Chatbot & Multi-Agent Integration	20%	Query accuracy, agent coordination, UX, error handling
Documentation & Presentation	10%	Clarity, completeness, demo quality
Code Quality & Best Practices	5%	Clean code, patterns, Git usage

7.1 Bonus Points (Up to 10%)

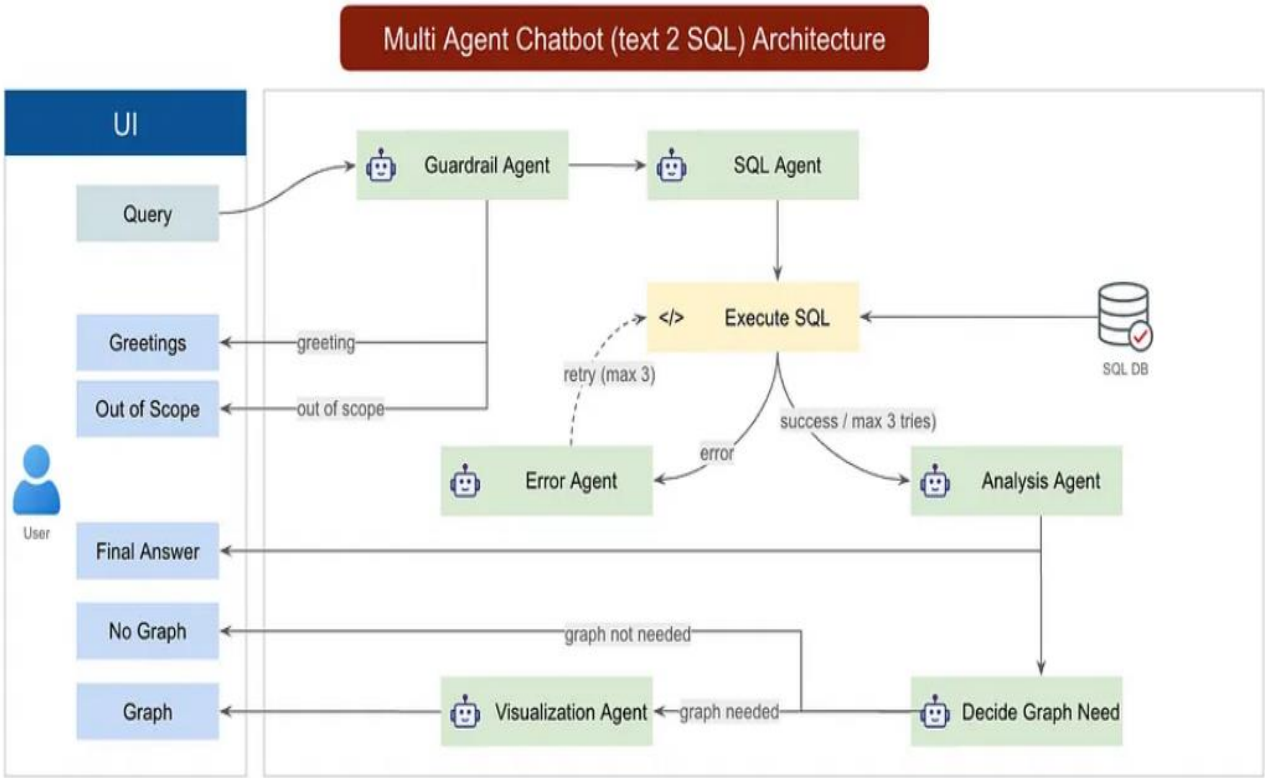
- Docker/Kubernetes deployment configuration
- Real-time notifications using WebSockets
- CI/CD pipeline implementation

8. Resources and References

8.1 Dataset Sources (Kaggle)

- UCI Online Retail Dataset (E-Commerce Sales Forecast)
- E-Commerce Customer Behavior Dataset
- E-Commerce Shipping Data
- E-Commerce Sales (Amazon)
- Pakistan's Largest E-Commerce Dataset
- Amazon US Customer Reviews Dataset

Multi Agent Chatbot Architecture



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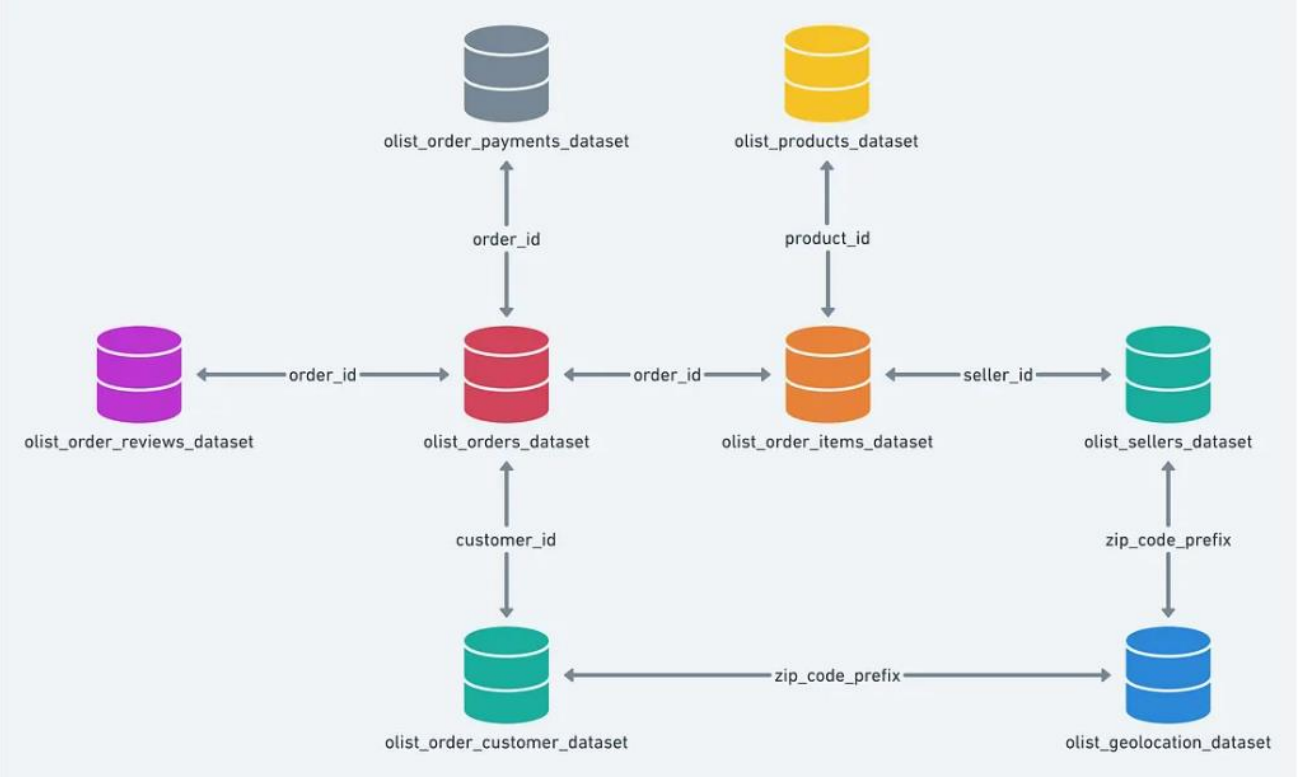
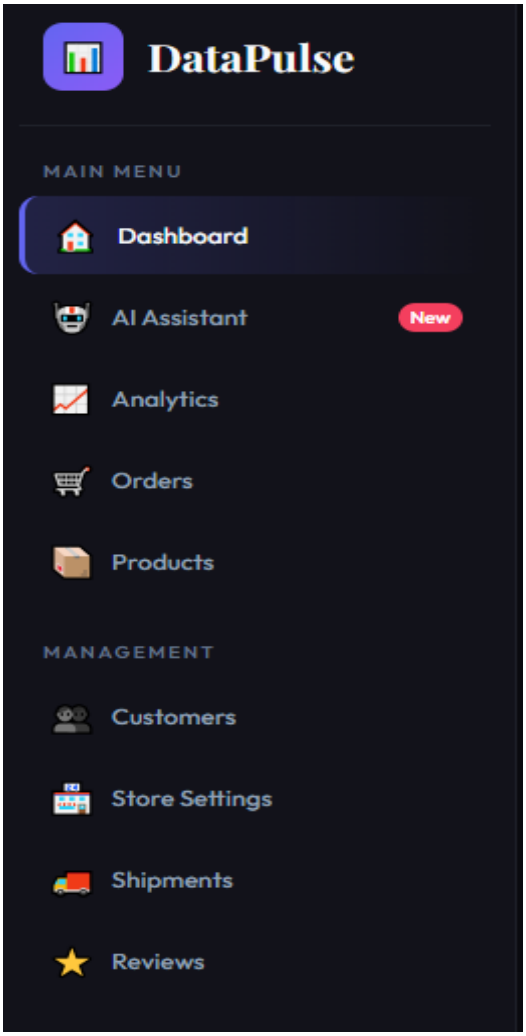
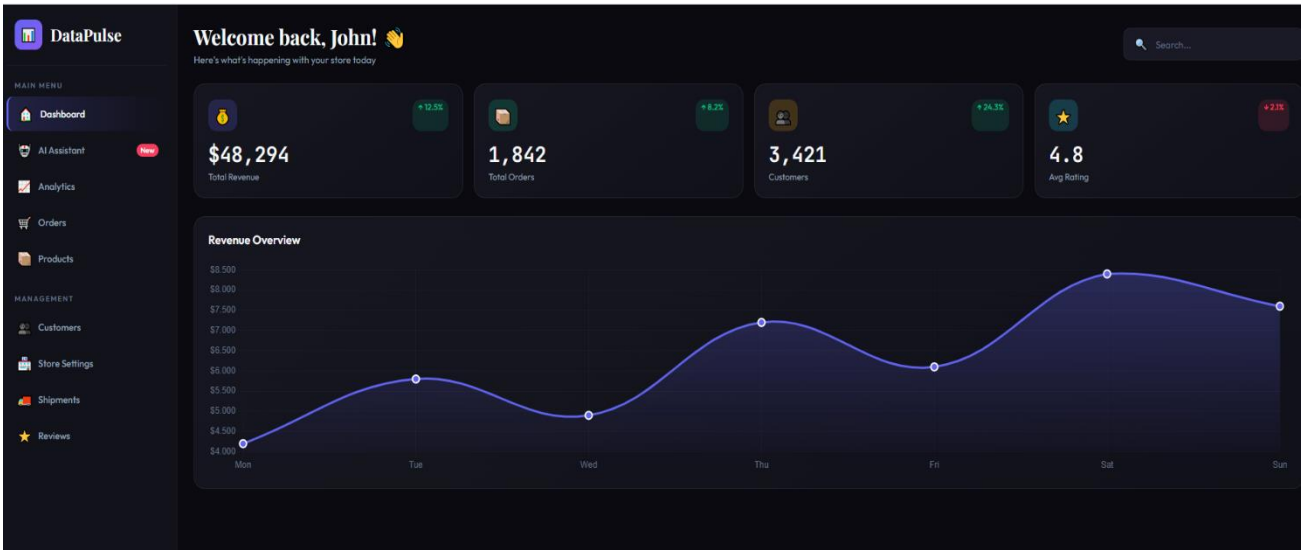


Example Screen





AI Data Assistant

Ask anything about your store data - I'll visualize it for you!

DataPulse AI

• Online - Ready to analyze

Hello John! 👋 I'm your AI data assistant with **real-time visualization** capabilities!

I can analyze your data and create charts instantly. Try asking:
- "Show sales by category"
- "Weekly revenue chart"
- "Top products performance"
- "Find negative reviews"
- "Customer distribution"

Show order status breakdown JD

Order Status Breakdown

Current order distribution:
- ✅ Completed: 1,437 (78%)
- 🚚 Shipped: 276 (15%)
- ⏳ Pending: 92 (5%)
- ❌ Cancelled: 37 (2%)

✅ 78% completion rate is excellent!

Top 5 products performance JD

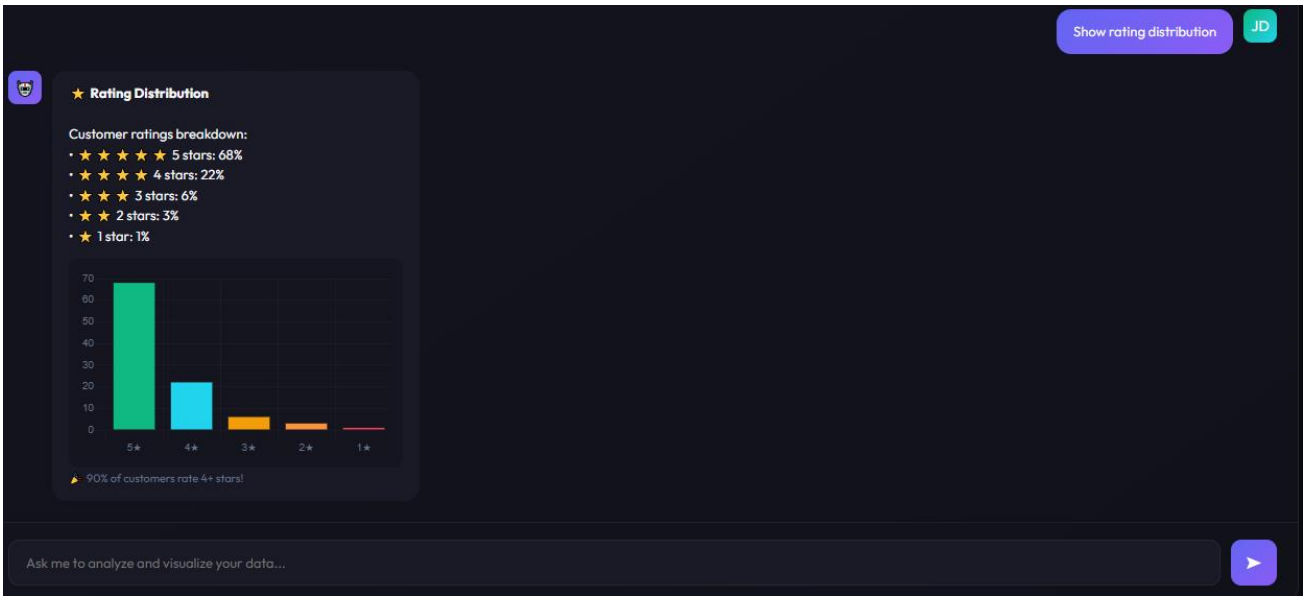
Show weekly revenue trend JD

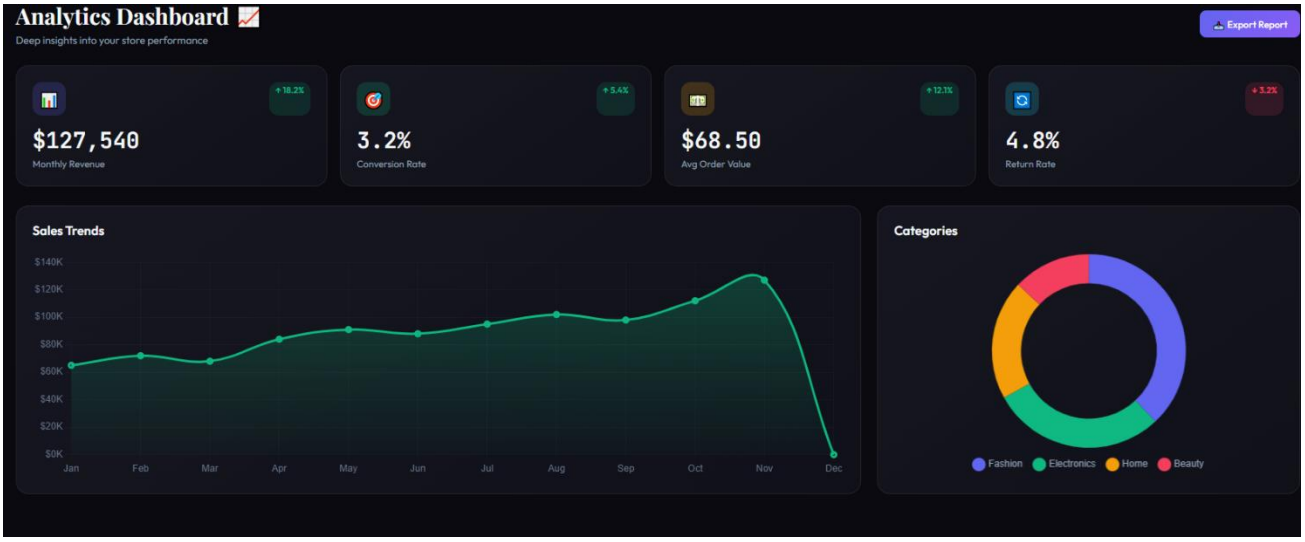
Weekly Revenue Trend

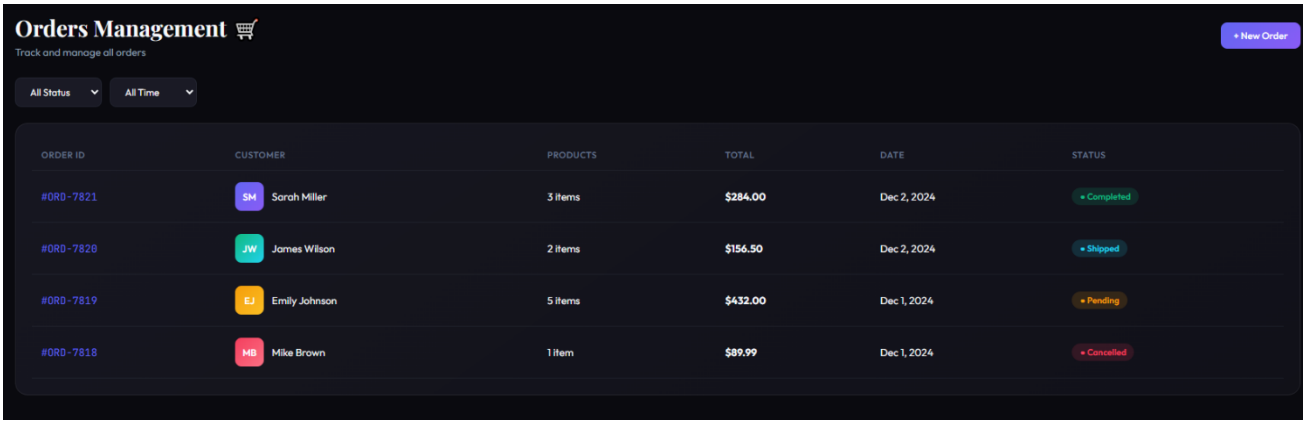
Your revenue performance this week:
- Peak day: Saturday ($8,400)
- Lowest: Tuesday ($4,200)
- Weekly total: $48,294

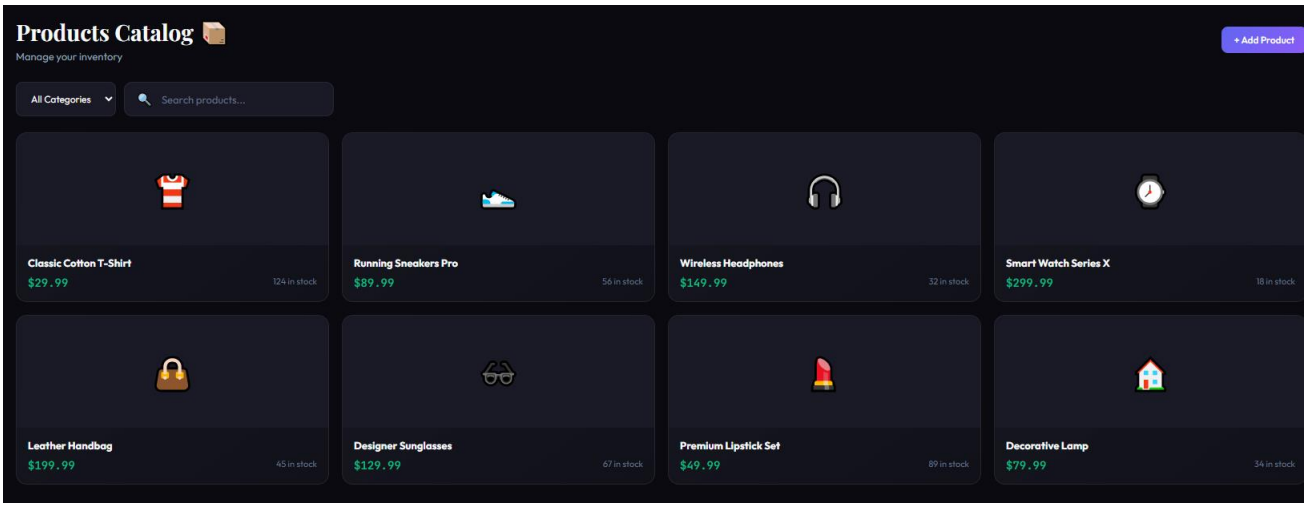
📈 Weekend sales are 45% higher!

Ask me to analyze and visualize your data...


Show rating distribution
JD
Rating Distribution
Customer ratings breakdown:
• ★ ★ ★ ★ ★ 5 stars: 68%
• ★ ★ ★ ★ 4 stars: 22%
• ★ ★ ★ 3 stars: 6%
• ★ ★ 2 stars: 3%
• ★ 1 star: 1%
70
60
50
40
30
20
10
0
5★
4★
3★
2★
1★
90% of customers rate 4+ stars!
Ask me to analyze and visualize your data...


Analytics Dashboard
Deep insights into your store performance
Export Report
+18.2%
$127,540
Monthly Revenue
+5.4%
3.2%
Conversion Rate
+12.1%
$68.50
Avg Order Value
+3.2%
4.8%
Return Rate
Sales Trends
$140K
$120K
$100K
$80K
$60K
$40K
$20K
$0K
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Categories
Fashion
Electronics
Home
Beauty


Orders Management
Track and manage all orders
+ New Order
All Status
All Time
ORDER ID
CUSTOMER
PRODUCTS
TOTAL
DATE
STATUS
#ORD-7821
SM
Sarah Miller
3 items
$284.00
Dec 2, 2024
Completed
#ORD-7820
JW
James Wilson
2 items
$156.50
Dec 2, 2024
Shipped
#ORD-7819
EJ
Emily Johnson
5 items
$432.00
Dec 1, 2024
Pending
#ORD-7818
MB
Mike Brown
1 item
$89.99
Dec 1, 2024
Cancelled


Products Catalog
Manage your inventory
+ Add Product
All Categories
Search products...
Classic Cotton T-Shirt
$29.99
124 in stock
Running Sneakers Pro
$89.99
56 in stock
Wireless Headphones
$149.99
32 in stock
Smart Watch Series X
$299.99
18 in stock
Leather Handbag
$199.99
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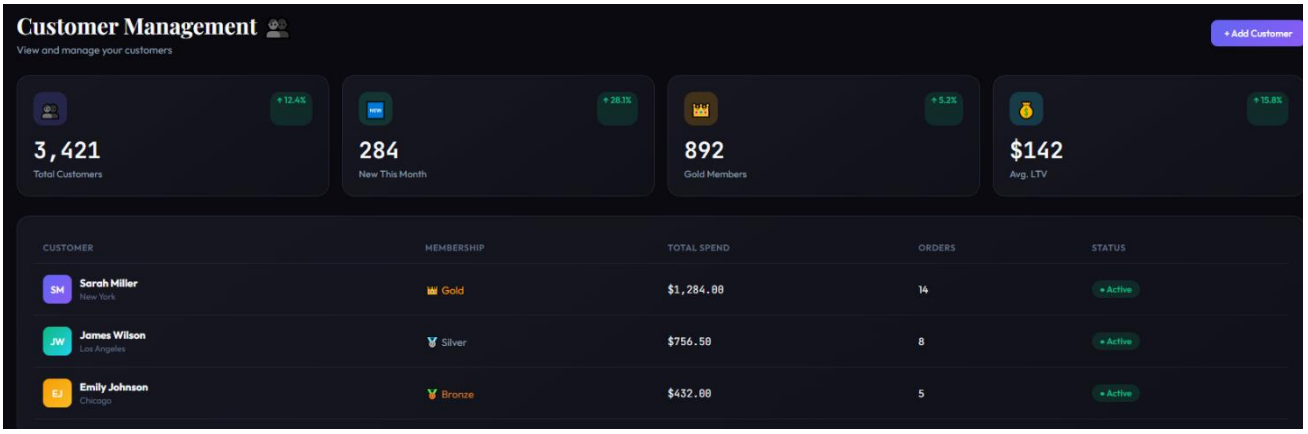
67 in stock
Premium Lipstick Set
$49.99
89 in stock
Decorative Lamp
$79.99
34 in stock


Customer Management
View and manage your customers
+ Add Customer
+ 12.4%
3,421
Total Customers
+ 28.1%
284
New This Month
+ 5.2%
892
Gold Members
+ 15.8%
$142
Avg. LTV
CUSTOMER
MEMBERSHIP
TOTAL SPEND
ORDERS
STATUS
SM
Sarah Miller
New York
Gold
$1,284.00
14
Active
JW
James Wilson
Los Angeles
Silver
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8
Active
EJ
Emily Johnson
Chicago
Bronze
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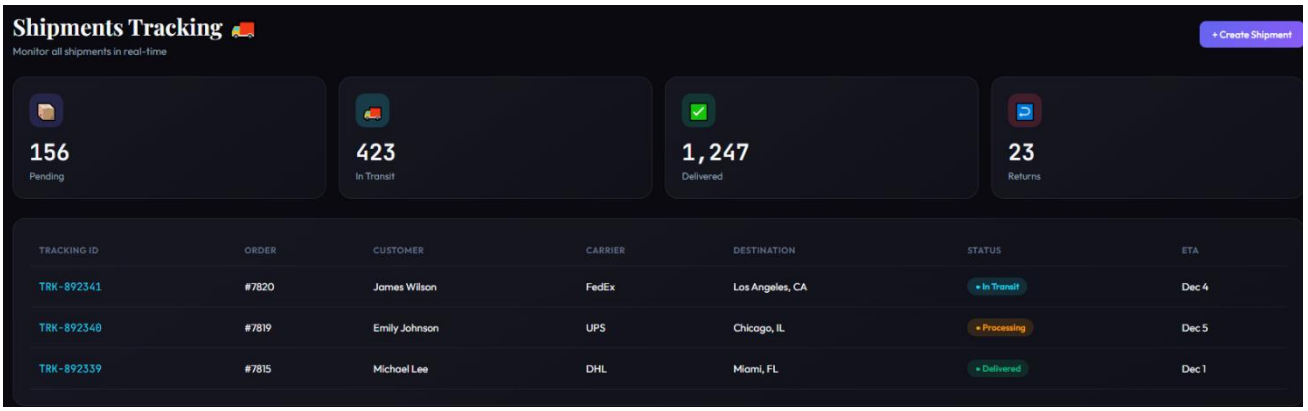
5
Active


Shipments Tracking
Monitor all shipments in real-time
+ Create Shipment
156
Pending
423
In Transit
1,247
Delivered
23
Returns
TRACKING ID
ORDER
CUSTOMER
CARRIER
DESTINATION
STATUS
ETA
TRK-892341
#7820
James Wilson
FedEx
Los Angeles, CA
In Transit
Dec 4
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Emily Johnson
UPS
Chicago, IL
Processing
Dec 5
TRK-892339
#7815
Michael Lee
DHL
Miami, FL
Delivered
Dec 1